CSM
Spring 2026

EECS 16B
Worksheet 4

1. Complex Number Practice

(a) Take z_1 and z_2 to have the following rectangular coordinates form:

$$z_1 = 2 + 6j$$
$$z_2 = 3 - 5j.$$

What is the real part of $z_1 + z_2$?

(b) What is the imaginary part of $z_1 + z_2$?

(c) How do we visualize the polar representation of complex numbers, $z = Ae^{j\theta}$, in the complex plane?

(d) Given $z = 7e^{j30^\circ}$, what is $|z|$?

(e) What is $\angle z$?

(f) Given $z = 5e^{j90^\circ}$, what is the real part of z?

(g) What is the imaginary part of z?

CSM, Spring 2026. All Rights Reserved. This may not be publicly shared without explicit permission.

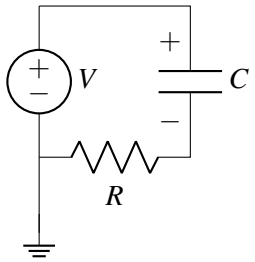
2. Introduction to Capacitors

In this problem, we will introduce the fundamental circuit components.

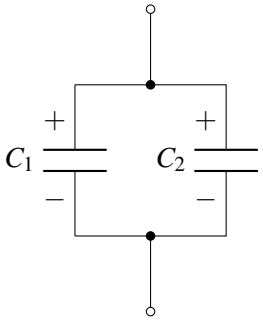
(a) What is a capacitor?

(b) What is capacitance?

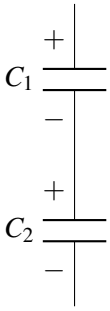
(c) What happens to the following circuit (specifically with respect to the voltage across and current through the capacitor) when
a) the capacitor becomes fully charged?
b) after the capacitor is fully charged, we remove the voltage source and replace it with a short?



(d) Using physical properties of capacitors, derive the equation to find the equivalent capacitance for two capacitors in parallel, C_1 and C_2 in the figure below. Use this equation to find a relationship between the total charge of the two capacitors and the charge of each individual capacitor.



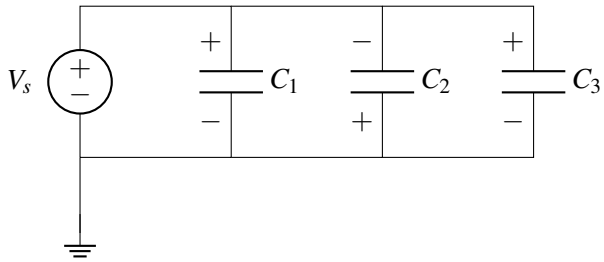
(e) Using physical properties of capacitors, derive the equation for the equivalent capacitance of two capacitors in series.



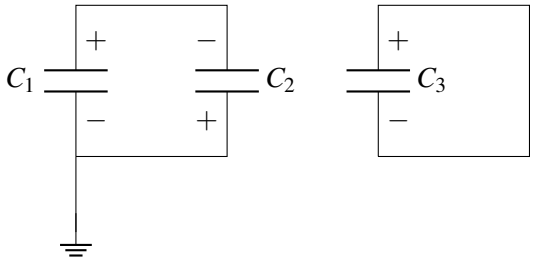
CSM, Spring 2026. All Rights Reserved. This may not be publicly shared without explicit permission.

3. Pitfall Problem

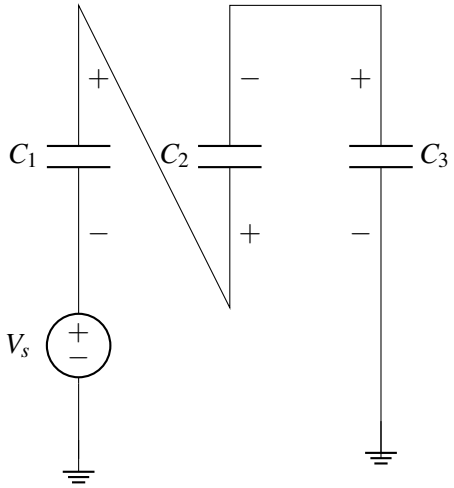
(a) Consider the following circuit in ϕ_1. Assume that all capacitors are initially discharged. Find out the charge on each capacitor in this phase.



(b) Assume that ϕ_1 has taken place, and that the capacitors are then moved to the following configuration in ϕ_2. Calculate the charge across each capacitor in ϕ_2.



(c) Assume that ϕ_2 has taken place, and that the capacitors are then moved to the following configuration in ϕ_3. Calculate the charge across each capacitor in ϕ_3.



Contributors:

- Sirej Dua.
- Yannan Tuo, Emily Gosti, Anwar Baroudi.
- Mudit Gupta.

CSM, Spring 2026. All Rights Reserved. This may not be publicly shared without explicit permission.